

Statistical Models for Discrete Outcomes

Practical 2 Solutions

Part A: Investigation into low birthweights in Massachusetts, USA

3.

```
. glm low lwt, family(binomial) link(logit)
```

Generalized linear models		Number of obs	=	189
Optimization	: ML	Residual df	=	187
		Scale parameter	=	1
Deviance	=	228.708058	(1/df) Deviance	= 1.223038
Pearson	=	189.640232	(1/df) Pearson	= 1.014119
Variance function: V(u) = u*(1-u)		[Bernoulli]		
Link function : g(u) = ln(u/(1-u))		[Logit]		
Log likelihood = -114.354029		AIC	=	1.23126
		BIC	=	-751.4986

		OIM				
	low	Coefficient	std. err.	z	P> z	[95% conf. interval]

	lwt	-.0140371	.0061685	-2.28	0.023	-.0261272 -.001947
	_cons	.9957634	.7852435	1.27	0.205	-.5432856 2.534812

Discuss: Interpret both of the estimated coefficients and their associated p-values and 95% confidence intervals (CI).

0.9957634: the estimated log odds of a low birthweight baby when mother's weight is zero (an extrapolation).

It can be converted to the estimated odds of a low birthweight baby when mother's weight is zero by taking its exponent ($e^{0.9957634} = 2.706788$).

This can be converted to the estimated probability of a low birthweight baby when mother's weight is zero using the standard conversion formula.

$$\text{probability} = \frac{\text{odds}}{\text{odds}+1}, \text{ obtained by rearranging } \text{odds} = \frac{\text{probability}}{1-\text{probability}} \quad (1)$$

Here this estimated probability is 0.730 ($2.706788/(2.706788 + 1)$). Since it is an extrapolation to an impossible mother's weight this (and its 95% CI) is meaningless from a practical perspective. It is reported in the Stata output because in order to define any straight line we require the slope of the line and one point on the line. Stata's default is to report the point where the covariate equals zero. Often this is meaningless as a standalone

value. It can sometimes be made more informative by centring continuous variables around their mean or median (or a value close to one of these).

The reported p-value for the constant term relates to a test of the null hypothesis that log odds of a low birthweight baby equals zero, *i.e.*, that the probability equals 0.5. There is no practical interest in this null hypothesis or the p-value.

-.0140371: the estimated increase in the log odds of a low birthweight baby per 1lb increase in mother's weight at last menstrual period.

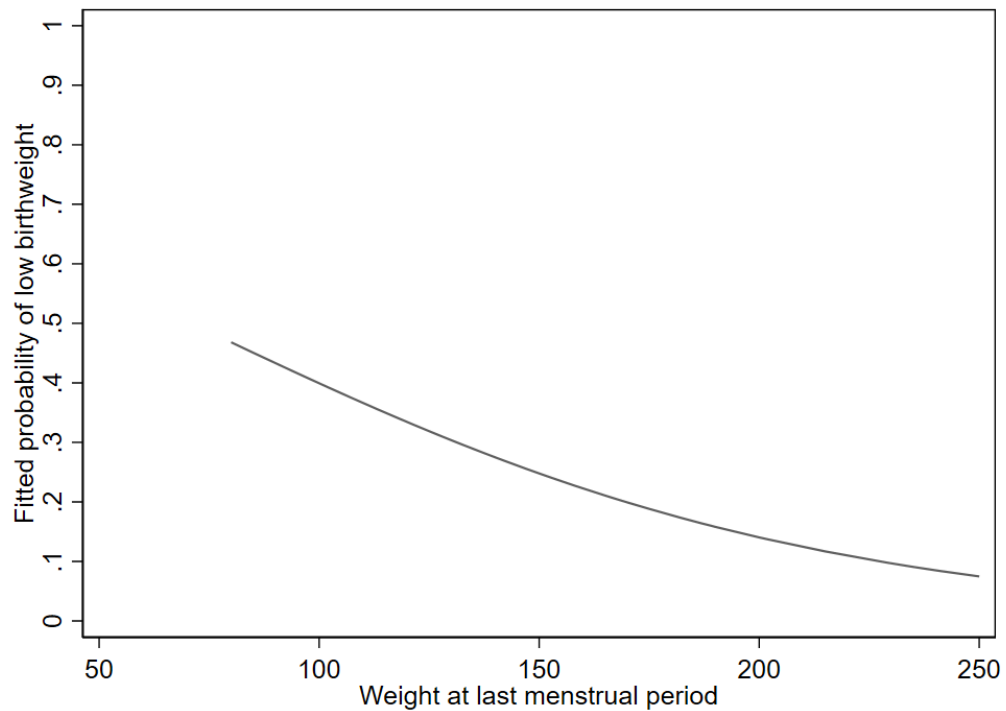
It can be converted to the estimated odds ratio (of having a low birthweight baby) per 1 lb increase in mother's weight by taking its exponent ($e^{-0.0140371} = 0.9860609$). Subtracting 1 from this estimate gives -0.0139391, so we might report that the odds of a low birthweight baby is reduced by an estimated 1.4% per 1lb increase in mother's weight at last menstrual period.

The reported p-value for the slope relates to a test of the null hypothesis that the log odds ratio per 1lb increase in mother's weight is zero, *i.e.*, that the odds ratio equals 1. Here we have evidence ($p=0.023$) that the odds (and therefore the probability) of a low birthweight baby is associated with mother's weight.

The reported 95% CI for the slope relates to the log odds ratio per 1lb increase in mother's weight. We would usually convert the confidence limits to confidence limits for the odds ratio by taking exponents ($e^{-0.0261272} = 0.9742112$, $e^{-0.001947} = 0.9980549$). The interpretation of the 95% CI is the standard one. We can either say that there is a 95% probability that a 95% CI constructed in this way has a 95% probability of containing the true (population) odds ratio, or that the results are consistent ($p>0.05$) with true odds ratios between 0.974 and 0.998 and inconsistent ($p<0.05$) with true odds ratios outside this range.

We could summarise the important results here as **"There is evidence ($p=0.023$) of an association between mother's weight and the probability of a low birthweight baby. Odds of a low birthweight baby were estimated to be reduced by 1.4% (95% CI 0.2% to 2.6%) per 1lb increase in mother's weight at last menstrual period"**.

We might also display the fitted probabilities of a low birthweight baby by mother's weight in a graph as shown overleaf (see the code to produce this in the Do file).



4. The following four commands can all be used to fit logistic regression models.

```
glm low lwt, fam(bin) link(logit)
logit low lwt
glm low lwt, fam(bin) link(logit) eform
logistic low lwt
```

The first two commands report estimates of the log odds (at mother's weight zero) and log odds ratios (for a 1 lb increase in mother's weight). The latter two convert these to odds and odds ratios respectively. There are differences in the other information that is reported too. We will consider these in later sessions. Note that the reported log-likelihood from the model is the same in each case. Note also that the commands differ somewhat with regards to what post estimation commands are available after fitting the model, something that we will be looking at in later sessions.

5. The estimated log odds of low birthweight for a mother who weighed 120 lbs is

$$(120 \times -0.014037) + 0.995763 = -0.6887.$$

Hence, the probability is

$$\exp(-0.6887) / (1 + \exp(-0.6887)) = 0.334 .$$

6. Using lincom:

```
lincom 120*lwt + _cons, eform
```

```
( 1) 120*[low]lwt + [low]_cons = 0
```

	low	exp(b)	Std. err.	z	P> z	[95% Conf. interval]
	(1)	.5022331	.0813097	-4.25	0.000	.365678 .6897819

This gives the estimated odds of low birthweight for a mother who weighed 120 lbs.

Rearranging using formula (1) shows that the estimated probability of low birthweight for a mother who weighed 120 lbs is

$$0.5022331 / (1 + 0.5022331) = 0.334 \text{ (as calculated by hand above).}$$

The lower limit of the 95% confidence interval for the probability can be calculated as:

$$0.365678 / (1 + 0.365678) = 0.268 .$$

Analogously, the upper limit is given by

$$0.6897819 / (1 + 0.6897819) = 0.408 .$$

Alternatively, re-parameterise the model:

```
. gen lwt120 = lwt - 120
. logistic low lwt120
```

Logistic regression	Number of obs	=	189
	LR chi2(1)	=	5.96
	Prob > chi2	=	0.0146
Log likelihood = -114.35403	Pseudo R2	=	0.0254

	low Odds Ratio	Std. err.	z	P> z	[95% conf. interval]	
lwt120	.9860609	.0060825	-2.28	0.023	.9742112	.9980549
_cons	.5022331	.0813097	-4.25	0.000	.365678	.6897819

Note: _cons estimates baseline odds.

Here, the constant term is the odds of low birthweight for a woman of 120 lbs; it is the same result as given by the lincom command above.

7.

```
. logit low i.race lwt
```

```
Logistic regression               Number of obs   =       189
                                LR chi2(3)         =       11.40
                                Prob > chi2        =       0.0098
Log likelihood = -111.63836       Pseudo R2      =       0.0486
```

```
-----+-----
      low | Coefficient   Std. err.      z    P>|z|     [95% Conf. interval]
-----+-----
      race |
    black |   1.080719    .4880135     2.21   0.027    .1242306    2.037208
    other |   .4810363    .3566464     1.35   0.177   -.2179778    1.18005
      _cons |  -.0152009    .0064381    -2.36   0.018   -.0278193   -.0025825
      _cons |   .8029594    .8450592     0.95   0.342   -.8533262    2.459245
-----+-----
```

```
. est store A
```

```
. logit low lwt
```

```
Logistic regression               Number of obs   =       189
                                LR chi2(1)         =        5.96
                                Prob > chi2        =       0.0146
Log likelihood = -114.35403       Pseudo R2      =       0.0254
```

```
-----+-----
      low | Coefficient   Std. err.      z    P>|z|     [95% Conf. interval]
-----+-----
      lwt |  -.0140371    .0061685    -2.28   0.023   -.0261272   -.001947
      _cons |   .9957626    .7852434     1.27   0.205   -.5432862    2.534811
-----+-----
```

```
. est store B
```

```
. lrtest A B
```

```
Likelihood-ratio test               LR chi2(2)   =        5.43
(Assumption: B nested in A)         Prob > chi2 =       0.0662
```

Discussion: What do you conclude?

Comparing the fits of these models tests the null hypothesis that race has no independent association with the probability of having a low birthweight baby, after adjusting for mother's weight. The log-likelihood ratio statistic for this comparison is $-2 \times (-114.35403 - -111.63836) = 5.43$ which is compared to a chi-square distribution with two degrees of freedom, giving $p = 0.066$. There is some weak evidence that race is independently associated with low birthweight, after adjusting for mother's weight. A Wald test of the same null hypothesis gives a very similar p-value here (see do file).

8. From the model with both mother's weight and race as covariates we have the following.

```
LR chi2(3) = 11.40
Prob > chi2 = 0.0098
```

The LR value and p-value match those from the profile likelihood ratio test comparing this model with the null model (no covariates). We can therefore conclude that the LR statistic and p-value displayed in the top right after fitting a logistic regression model corresponds to the likelihood ratio test of the null hypothesis that the coefficients corresponding to the covariates included in the non-null model are all zero. That is, it is testing whether the outcome/dependent variable is independent of all the included covariates. It is analogous to the overall F-test from a linear regression model.

Note that this test is often not very useful in practice. In this example all we can conclude from this test is that there is evidence that mother's weight and/or race are associated with the odds of a low birthweight baby. We cannot conclude that there is evidence that both factors are associated with the outcome, only that there is evidence that at least one of them is.

Part B: Investigation into effect of CS₂ at different doses on insect survival

9.

```
. table dose
```

	Frequency
Dose (mg/l)	
49.06	59
52.99	60
56.91	62
60.84	56
64.76	63
68.69	59
72.61	62
76.54	60
Total	481

The dataset contains information on 481 insects, with approximately 60 insects in each of the dose groups.

10.

```
. glm y dose, family(binomial) link(logit)
```

Generalized linear models		Number of obs	=	481
Optimization	: ML	Residual df	=	479
		Scale parameter	=	1
Deviance	= 383.4586582	(1/df) Deviance	=	.80054
Pearson	= 478.0602265	(1/df) Pearson	=	.9980381

Variance function: $V(u) = u*(1-u)$	[Bernoulli]
Link function : $g(u) = \ln(u/(1-u))$	[Logit]

Log likelihood	= -191.7293291	AIC	=	.8055274
		BIC	=	-2574.782


```
-----+-----
```

		OIM				
y	Coefficient	std. err.	z	P> z	[95% conf. interval]	
dose	.2365929	.0203032	11.65	0.000	.1967994	.2763864
_cons	-14.0864	1.228393	-11.47	0.000	-16.49401	-11.6788

```
-----+-----
```



```
. est store A
. predict p1
(option mu assumed; predicted mean y)
```

The Wald test of the linear term has a z-score of 11.65, and $p < 0.001$. So strong evidence of increasing risk of death with increasing dose. Note that the final two lines of code shown here store the i) log-likelihood and ii) predicted probabilities of death for later use.

11.

```
. glm y dose, family(binomial) link(logit) eform
```



```
Log likelihood = -191.7293291
```



```
-----+-----
```

		OIM				
y	Odds ratio	std. err.	z	P> z	[95% conf. interval]	
dose	1.266925	.0257226	11.65	0.000	1.2175	1.318357
_cons	7.63e-07	9.37e-07	-11.47	0.000	6.87e-08	8.47e-06

```
-----+-----
```

Note: _cons estimates baseline odds.

There is strong evidence that the probability of death increases with increasing CS₂ dose ($p < 0.001$). Odds of death are estimated to increase by 27% (95% confidence interval 22% to 32%) per unit mg/l increase in dose.

12.

The estimated probability of death at a dose of 55 mg/l is given by the following.

$$\text{logit}(\hat{\pi}_i) = -14.09 + (0.2366 \times 55) = -1.077$$

$$\text{So } \hat{\pi}_i = \frac{e^{-1.077}}{1 + e^{-1.077}} = 0.254.$$

See the Do file for how to use `lincom` to obtain a 95% CI for this estimated probability.

13.

The dose that, on the basis of this model, would lead to a 50% death rate (LD50) is the dose that is associated with $\pi = 0.5$ and hence $\text{logit}(\pi) = 0$. This can be found by solving the following equation.

$$0 = -14.09 + (0.2366 \times \text{LD50})$$

Hence LD50 = 59.5 mg/l.

14.

```
. gen dose2 = dose^2
. glm y dose dose2, fam(binomial) link(logit)
```

```
Log likelihood   = -191.0133912
```

		OIM				
	y	Coefficient	std. err.	z	P> z	[95% conf. interval]
dose		-.1500052	.3291798	-0.46	0.649	-.7951857 .4951753
dose2		.003187	.0027273	1.17	0.243	-.0021584 .0085325
_cons		-2.488177	9.867752	-0.25	0.801	-21.82862 16.85226

Note: `_cons` estimates baseline odds.

```
. est store B
. predict p2
. lrtest A B
```

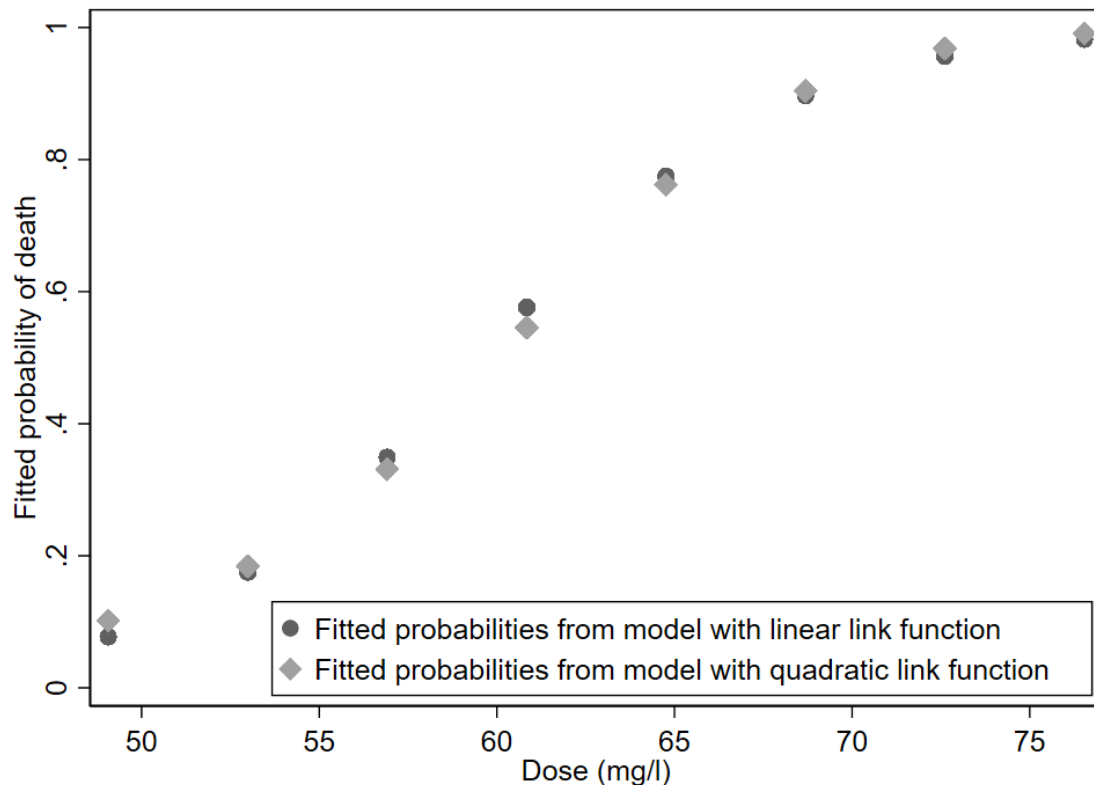
```
Likelihood-ratio test
Assumption: A nested within B
```

```
LR chi2(1) = 1.43
Prob > chi2 = 0.2315
```

The likelihood ratio test comparing the two models gives $p = 0.2315$, indicating no evidence against the null hypothesis that the linear dose model is correctly specified. There is thus no evidence of a quadratic effect of dose. The Wald test p -value for the same null hypothesis is $p = 0.243$. They are very close, and typically they will be, since the tests are asymptotically equivalent.

Note that neither the linear or quadratic terms are statistically significant in the above model. This does not mean that we have no evidence of an association (indeed we have already seen that there is strong evidence of a linear association). One of these terms (but not both) can be removed from the model without making the fit worse. In the interests of simplicity we would usually drop the quadratic term.

We can also display the fitted probabilities of death given by the two models (see the code to produce this graph in the Do file). Here the two predictions are close at all dose levels.



Note that for each model the predicted probabilities asymptotically approach 1 as the dose increases (and asymptotically approach 0 as the dose decreases).

15.

```
. glm y i.dosegroup, family(binomial) link(logit)
```

```
Log likelihood   = -189.4215899
```

		OIM				
y	Coefficient	std. err.	z	P> z	[95% conf. interval]	
dosegroup						
2	.8933342	.5326671	1.68	0.094	-.1506742	1.937343
3	1.284715	.5136316	2.50	0.012	.2780151	2.291414
4	2.178532	.5069153	4.30	0.000	1.184997	3.172068
5	3.731881	.5437596	6.86	0.000	2.666132	4.79763
6	4.357065	.6091545	7.15	0.000	3.163144	5.550986
7	5.57973	.8379745	6.66	0.000	3.93733	7.22213
8	6.256069	1.096578	5.71	0.000	4.106816	8.405322
_cons	-2.178532	.4307373	-5.06	0.000	-3.022762	-1.334303

The constant is -2.179. This is the log odds of death in the baseline category (Group 1). Hence the probability is $e^{-2.179}/(1 + e^{-2.179}) = 0.10$.

The coefficient for the second dose category is 0.893. This is the log odds ratio comparing dose category 2 with category 1. Hence the log odds of death in category 2 is $-2.179 + 0.893 = -1.285$. It follows that the probability of death in group 2 is $e^{-1.285}/(1 + e^{-1.285}) = 0.22$.

Interpretations for the other parameter estimates are analogous to this.

Discussion: Explain why the earlier model where dose was treated as a continuous predictor is nested in this one.

First note that in this “categorical” dose model the predicted probabilities are the same as the empirically observed probabilities in each dose group. This is because dose only takes eight distinct values, so a model with eight parameters will give predicted probabilities that exactly match the observed group-specific probabilities. So, the model is effectively estimating the eight dose group probabilities of death with no constraints on these. The model with dose as continuous also gives just eight predictions (one for each dose) for the data at hand, but now the predicted probabilities are constrained such that they follow a linear (on the logit scale) relationship with dose. Hence this “continuous” dose model is nested in the “categorical” dose model.

This can be seen empirically by fitting a model where dose is treated as both continuous and categorical (see the Do file). To allow dose to be included as continuous only six dose group indicator variables are included and estimated by Stata. The log-likelihood (-189.4215899) and predictions from this model are seen to be identical to those for the model with dose as categorical.

Note that models where a predictor is treated as continuous are not nested in models where that predictor is categorised by grouping together similar (but not identical) levels of that predictor. For example, models where we i) consider age as continuous and ii) categorise age as “20-29”, “30-39” etc. are not nested.

16.

```
. est store C  
. lrtest A C
```

Likelihood-ratio test
Assumption: A nested within C

```
LR chi2(6) = 4.62  
Prob > chi2 = 0.5940
```

We have no evidence that the “categorical” dose model fits the data better than the “continuous” dose model. Predicted probabilities from the two models are not markedly different (see the Do file for the code to produce this graph).

